A BEGINNER'S GUIDE

GenAI Terms, Explained Simply

From AI to Agentic AI, told as one connected story. Sixteen terms, each with a plain-English explanation, an analogy, and a real-life example.

How to read this guide

Every term in this guide is explained through **one student**. We follow him from a big idea (an artificial student who can think), through his brain, his reading habits, his school years and his college training, to his first job, and finally to a whole team of these students running a project. Each section ends with a short *story link* that leads into the next term, so the whole guide reads like one journey instead of a dictionary.

Part	What it answers	Terms
1. The Big Picture	What is AI, and how does it read?	AI, Model, LLM, Transformer, Token, Context Window, Multimodal
2. How It Learns	How does a model understand and get built?	Embeddings, Parameters, Training, Pre-training, Fine-tuning, Hallucination
3. From Answers to Actions	How do the four big AI words differ?	AI, Gen AI, AI Agent, Agentic AI

Tip: read the parts in order the first time. Later, use the summary tables at the end of each part as quick revision notes.

PART 1

The Big Picture

We start with the biggest idea, AI, and zoom in step by step until we reach a model that can read, understand and answer in human language.

1 AI (Artificial Intelligence)

Machines that do things that normally need human intelligence.

AI is the broad field of building machines that can think, learn and make decisions in ways that normally need a human brain. It is not one product or one app. It is an umbrella that covers everything from the simple tools on your phone to the chatbots making headlines today.

Most older AI systems were built to **analyze, classify or predict**: is this email spam, is this face the phone owner, which route is fastest. They are smart, but they do not write or draw anything new.

> **THE STUDENT ANALOGY**
> AI is the dream of building an **artificial student** who can think and learn. At this point we are only describing the dream, not yet the brain that makes it real.

> **REAL-LIFE EXAMPLE**
> A spam filter moving junk out of your inbox, face unlock on your phone, Google Maps picking the fastest route, and Netflix suggesting your next show are all AI.

Story link: But how does a machine actually do all this? It needs a trained brain. That brain is called a model.

2 Model

The trained brain that takes your question and gives an answer.

A model is a system that has been **trained on data** so it has learned patterns. Before training it is empty and useless. After training, you can give it an input (a question, a photo, a sentence) and it produces an output (an answer, a label, a translation).

People often confuse the app with the model. The chat window you type into is only the app. The model is the brain working behind it.

> **THE STUDENT ANALOGY**
> The model is the **student's brain after he has learned**. You ask him a question, and whatever he has stored in his brain comes out as the answer.

> **REAL-LIFE EXAMPLE**
> ChatGPT, Claude and Gemini are all models. A model that spots cats in photos, or one that predicts house prices, is also a model.

Story link: Models come in many kinds. Some are built especially for language, and those get their own name.

3 LLM (Large Language Model)

A model that is an expert in human language.

An LLM is a model trained on a huge amount of text: books, articles, websites, code and more. **Large** refers to the enormous amount of data it learns from and the billions of internal values it carries. **Language** means it works with words.

At its core, an LLM learns to predict the next word in a sentence. Do that billions of times and it becomes surprisingly good at writing, summarizing, translating, answering questions and even writing code.

> **THE STUDENT ANALOGY**
> The LLM is a **student who has read millions of books**. Because of all that reading, he speaks and writes fluently, even though he never lived most of what he read about.

> **REAL-LIFE EXAMPLE**
> Ask it to summarize a five-page report, write a polite leave letter, explain a tax term in simple words, or fix a bug in your code. All of these are language tasks an LLM handles.

Story link: But how does it understand language so well? The secret is a special design inside it.

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4 Transformer

The design that helps the model focus on the words that matter.

The transformer is the **architecture**, the blueprint, inside almost every modern LLM. It was introduced in a 2017 research paper called *Attention Is All You Need*, and it changed the field.

Its key idea is **attention**: instead of reading one word at a time and forgetting the earlier ones, the model looks at the whole sentence at once and works out which words are most related to each other. That is also what lets it learn from huge amounts of text quickly.

> **THE STUDENT ANALOGY**
> Transformer is **how the student's brain pays attention**. In a noisy classroom he locks onto the teacher's voice and ignores the rest.

> **REAL-LIFE EXAMPLE**
> In the sentence *"The dog chased the cat because it was scared,"* a transformer works out that "it" means the cat, not the dog. Without attention, this simple task would confuse the model.

Story link: But before the model can focus on words, it first has to read them. It does that in small pieces.

5 Token

The small pieces of text that a model actually reads.

A model does not read sentences the way you do. It first chops text into small chunks called **tokens**. A token can be a whole word, part of a word, or even a punctuation mark. For example, a long word like "unbelievable" may be split into a few tokens.

As a rule of thumb in English, **100 tokens is about 75 words**. Tokens matter in practice because AI tools measure length, speed and cost in tokens, not in words.

> **THE STUDENT ANALOGY**
> The student reads **syllable by syllable**, like a child sounding out a new word. Each small piece he reads is one token.

> **REAL-LIFE EXAMPLE**
> The sentence "I love AI" is only a few tokens. Companies that offer AI through an API charge by the number of tokens you send and receive, so a longer prompt costs more.

Story link: But a model can only handle so many tokens at one time. That limit has its own name.

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6 Context Window

How much the model can keep in view at once.

The context window is the **maximum number of tokens** a model can handle in a single go. It includes everything: your messages, any document you paste, the earlier chat, and the model's own reply.

Early models had windows of only a few thousand tokens. Newer models can hold hundreds of thousands. But when the limit is crossed, the oldest part falls out and the model can no longer see it.

> **THE STUDENT ANALOGY**
> The context window is the **size of the student's desk**. A small desk fits one page and a big desk fits an entire book. If you keep piling pages on, the oldest ones slide off the edge and he forgets them.

> **REAL-LIFE EXAMPLE**
> In a very long chat, the AI forgets the name or instruction you gave at the start. Uploading a 300-page PDF needs a model with a large window. If a chat gets too long, start a new one with a short summary.

Story link: So far, everything we discussed was about text. But today's models go beyond words.

7 Multimodal

A model that understands more than text.

A **modality** is a type of data: text, images, audio or video. A multimodal model can take in, and often produce, more than one of these. Early language models only understood text. Newer models can look at a photo, listen to your voice or watch a short video and respond to it.

This is a big deal because the real world is not made of text alone. When a model can see and hear, it can help with far more of your daily problems.

> **THE STUDENT ANALOGY**
> The student is no longer limited to books. Now he can **read, look at pictures, listen to lectures and watch videos**, just like a student with all his senses working.

> **REAL-LIFE EXAMPLE**
> Take a photo of your fridge and ask, "What can I cook with this?" Screenshot an error message and ask for the fix. Photograph a handwritten note and ask for it to be typed out.

Story link: Whatever the input is, the model must turn it into numbers to understand it. That brings us to Part 2.

Part 1 at a glance

Term	In one line	Student analogy
AI	Machines that think and decide like humans	The dream of an artificial student
Model	The trained system that gives answers	His brain after learning
LLM	A model that is an expert in language	A student who read millions of books
Transformer	The design that focuses on important words	How his brain pays attention
Token	A small piece of text the model reads	Reading syllable by syllable
Context Window	How many tokens fit at once	The size of his desk
Multimodal	Understands text, images, audio, video	Using all his senses

PART 2

How It Learns

In Part 1 we saw that the model reads text as tokens. Now we look inside: how those tokens turn into meaning, where knowledge is stored, and how a model is built and trained.

8 Embeddings

Turning tokens into numbers that carry meaning.

A model cannot understand words. It can only work with numbers. So every token is converted into a long list of numbers called an **embedding**. These numbers are not random: they are arranged so that **similar meanings get similar numbers**.

Think of each embedding as an address on a giant map of meaning. Words about similar things live close to each other, and unrelated words live far apart.

> **THE STUDENT ANALOGY**
> Embeddings are **a map in the student's mind** where similar ideas sit close together. "Dog" and "puppy" are neighbors, while "dog" and "airplane" are far apart.

> **REAL-LIFE EXAMPLE**
> Search for "cheap flights" and a smart search also finds pages about "low-cost airfare," even though the words differ. Embeddings power this kind of search by meaning, as well as recommendations.

Story link: But how does the model know which numbers are right? Those values live inside the model.

9 Parameters

The internal values where a model stores what it has learned.

Parameters are the **adjustable values inside a model**, sometimes called weights. Each one is just a number, but together they hold everything the model has learned. Modern models have billions of them, which is why you see names like "7B" or "70B" (7 or 70 billion parameters).

More parameters generally means more capacity to learn, but also more computing power and cost. A bigger model is not always a better one for your task.

> **THE STUDENT ANALOGY**
> Parameters are the **billions of tiny connections between the student's brain cells**. The strength of each connection decides what he remembers and how he thinks.

> **REAL-LIFE EXAMPLE**
> When you read "Llama 70B," it means the model has about 70 billion of these values. A small model on your phone might have only a few billion.

Story link: But who sets these billions of values? Nobody does by hand. They are learned through training.

10 Training

How a model learns by practice and correction.

Training is the process of **adjusting the parameters** until the model gives good answers. It repeats a simple loop: show an example, let the model make a prediction, compare it with the correct answer, measure the error, and nudge the parameters to do slightly better next time.

This loop runs billions of times. It needs a huge amount of data, powerful chips called GPUs, and a lot of time and money.

> **THE STUDENT ANALOGY**
> Training is **going to school**. The student practices, makes mistakes, gets corrected, and slowly improves.

> **REAL-LIFE EXAMPLE**
> For the sentence "The capital of France is ___," the model first guesses "London." It is told the answer is "Paris," so its parameters shift a little. After countless such corrections, it gets it right.

Story link: But training happens in two stages, and the first one comes with a name of its own.

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11 Pre-training

The first, general stage of training.

Pre-training is the big first round. The model learns from a massive and varied collection of general data: books, websites, articles and code. It picks up grammar, facts, reasoning patterns and coding skills.

The result is a **base model** (also called a foundation model). It knows a lot about almost everything, but it is not yet specialized. It is also the most expensive stage, which is why only a few companies do it from scratch.

> **THE STUDENT ANALOGY**
> Pre-training is the student's **school years**, when he learns a bit of everything: language, science, history and general knowledge.

> **REAL-LIFE EXAMPLE**
> A base model has read plenty of medical text, but it is not yet ready to act as a careful assistant for your hospital. It has broad knowledge, not expertise.

Story link: So how do we turn a student with broad knowledge into an expert?

12 Fine-tuning

Extra training that makes a model an expert in one area.

Fine-tuning takes a pre-trained model and trains it further on a **smaller, specific set of data**. The parameters are adjusted again, this time to match a particular field, tone or task. It is far cheaper than pre-training, and methods such as LoRA make it cheaper still.

A useful tip: fine-tuning is best for changing how a model *behaves or writes*. For adding fresh facts that change often, connecting the model to your documents (a technique called RAG) is usually the better choice.

> **THE STUDENT ANALOGY**
> Fine-tuning is **college or job training**. The student already knows the basics from school, and now he specializes as a doctor, a lawyer, or your company's support agent.

> **REAL-LIFE EXAMPLE**
> Train a model on your company's past support chats and it starts answering customers in your brand's tone and format. Train on legal documents and it handles legal language better.

Story link: But even after all this training, the model is still not perfect.

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13 Hallucination

When a model confidently gives a wrong or made-up answer.

A hallucination is an answer that sounds right but is false. It happens because the model does not truly *know* facts. It predicts the most likely next words based on patterns stored in its parameters. When it lacks the right information, it rarely says "I don't know." It fills the gap with something that sounds believable.

Hallucinations are one of the biggest risks when building real solutions, especially in areas like health, law and finance.

> **THE STUDENT ANALOGY**
> In an exam, the student does not know the answer, but instead of admitting it he **bluffs with full confidence**. It sounds right, but it is wrong.

> **REAL-LIFE EXAMPLE**
> An AI may invent a book that does not exist, quote a fake legal case, give a wrong date, or call a code function that was never written. Lawyers have even been penalized for submitting AI-invented case citations.

> **HOW TO REDUCE IT**
> Verify important answers. Ask the model to cite sources. Give it the real documents to work from (RAG). Tell it to say "I don't know" when unsure. Lower the temperature setting for more predictable output.

Story link: AI makes mistakes, yet it keeps getting more useful, and now it does not only talk. It takes action. That created four terms people constantly mix up.

PART 3

From Answers to Actions

AI, Gen AI, AI Agent and Agentic AI sound alike but mean different things. Think of them as four circles, each one inside the one before it.

> **LEVEL ONE: AI (RECAP)**
> AI is the big umbrella, and everything below sits inside it. Most traditional AI, like spam filters, face unlock and Google Maps, only **analyzes or predicts**. It does not create anything new. That is where Gen AI comes in.

14 Gen AI (Generative AI)

Level two: AI that creates brand-new content.

Gen AI is a **part of AI**, not a separate thing. It is AI that can generate new text, images, music, video and code. LLMs are the best-known example, along with models that turn words into pictures.

Its limit: it gives you the output, but **you still take the action**. It can write a travel plan, but you have to book everything yourself.

> **THE STUDENT ANALOGY**
> Now the student does not only answer questions. He can **write an essay, paint a picture or compose a song** from scratch.

> **REAL-LIFE EXAMPLE**
> ChatGPT writing your email, or an app turning your sentence into an image, is Gen AI.

Story link: So the next step was to let Gen AI act, not just create.

15 AI Agent

Level three: Gen AI with tools that does a task for you.

An AI agent uses a Gen AI model as its **brain** and connects it to **tools** such as web search, email, calendar, payments or databases. It works in a loop: understand the goal, decide the next step, use a tool, check the result, and repeat until the task is done.

Agents are usually built for one job or a limited area, for example a travel assistant or a support assistant.

> **THE STUDENT ANALOGY**
> The student gets a **job, a laptop and a phone**. You say "book my flight," and he does not just explain how. He actually does it.

> **REAL-LIFE EXAMPLE**
> You say, "Book me a flight to Goa." The agent searches flights, compares prices, picks the best one, books it and adds it to your calendar. A support agent can look up your order and issue a refund.

Story link: But what if the goal is much bigger than one task?

16 Agentic AI

Level four: agents working together toward a big goal.

Agentic AI is the bigger system. One or more agents **plan the steps, divide the work, make decisions and adjust when something goes wrong**, with very little human help. The key difference is the size of the goal: not a single task, but a whole outcome.

More freedom also means more risk. The more autonomy a system has, the more it needs guardrails and human approval at important steps.

> **THE STUDENT ANALOGY**
> A **whole team of these employees** running a project, with a manager who splits the work and checks the result.

> **REAL-LIFE EXAMPLE**
> Goal: "Launch a marketing campaign." One agent researches the audience, one writes the content, one designs the visuals and one reviews everything before it goes out.

THE FINAL CHEAT SHEET

Four Terms, One Ladder

Each term is one step above the one before it. AI thinks, Gen AI creates, an agent acts, and Agentic AI achieves goals.

Term	What it does	Student analogy	Example
AI	Thinks and predicts	A student who can sort and decide	Spam filter
Gen AI	Creates new content	He writes, paints, composes	ChatGPT writing an email
AI Agent	Acts using tools	He gets a job and a laptop	Books your flight
Agentic AI	Achieves big goals	A team running a project	Runs a full campaign

The whole story in one paragraph

It all starts with **AI**, the idea of machines that think. A **model** is the trained brain that makes it real, and an **LLM** is a model that has read huge amounts of text. A **transformer** helps it focus on the right words. It reads in **tokens**, can hold only so many at once in its **context window**, and **multimodal** models extend this beyond text. Inside, **embeddings** turn tokens into meaning, and **parameters** store what was learned. **Training** fills those parameters: first **pre-training** for broad knowledge, then **fine-tuning** for expertise. Even then, the model can **hallucinate**. Finally, **Gen AI** creates, an **AI agent** acts, and **Agentic AI** pursues whole goals.

> **BEFORE YOU BUILD**
> Smart answers: a model is enough. Your own data: embeddings and RAG. Tasks: build an agent. Whole workflows: agentic systems, with guardrails and human checks.